
STATISTICAL PROCESS CONTROL

Analysis of a Semiconductor Manufacturing Process

Using the SECOM Industrial Dataset from UCI

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March 2026

Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy) · Microsoft Excel

Abstract

This report presents a Statistical Process Control (SPC) analysis of the SECOM semiconductor manufacturing dataset, a publicly available real-world industrial dataset from the UCI Machine Learning Repository. The dataset comprises measurements from **1,567 wafers** across **590 process parameters**, with a binary pass/fail quality label for each unit produced.

The analysis follows a structured five-phase methodology: data cleaning and preparation, feature selection, SPC chart construction, process capability analysis, and reporting. Python was used as the primary analytical tool, with Microsoft Excel for results reporting. Key findings include significant process instability in Parameter 59 — the parameter most correlated with wafer failures — with **37 out-of-control subgroups** on the X-bar chart and **32 on the R chart**, concentrated heavily in the first 500 wafers of production before the process stabilised.

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1. Introduction

1.1. Background

Semiconductor manufacturing is one of the most precision-demanding industrial processes in the world. Wafer fabrication involves hundreds of tightly controlled process steps, and even minor deviations in parameters such as temperature, pressure, or chemical flow rate can result in defective products. With failure rates as low as 5–8% considered acceptable in high-volume production environments, the ability to detect and diagnose process instability early is critical to maintaining yield and reducing waste.

Statistical Process Control (SPC) provides a systematic, data-driven framework for monitoring process behaviour over time. By distinguishing between *common cause variation* (inherent, expected variation) and *special cause variation* (unusual, assignable events), SPC enables engineers to intervene at the right time — neither over-adjusting a stable process nor ignoring genuine signals of deterioration.

1.2. Dataset Description

The SECOM dataset was sourced from the UCI Machine Learning Repository and represents production data from a real semiconductor fabrication plant. Key characteristics are summarised in Table 1.

Table 1: SECOM Dataset Overview

Attribute	Detail
Source	UCI Machine Learning Repository
Total wafers	1,567
Process parameters	590
Pass (label = -1)	1,463 (93.4%)
Fail (label = +1)	104 (6.6%)
Missing values	Present in 538 of 590 columns
File format	Space-separated .data files

1.3. Objectives

The objectives of this analysis are:

1. Clean and prepare the raw dataset for SPC analysis.
2. Reduce 590 parameters to a meaningful subset using variance filtering and correlation analysis.
3. Construct X-bar, R, and CUSUM control charts for the most critical parameters.
4. Calculate process capability indices (C_p , C_{pk}) and assess process performance.
5. Summarise actionable findings in the form of a quality engineering report.

1.4. Tools and Technologies

Table 2: Tools Used in This Analysis

Tool	Purpose	Libraries / Version
Python 3.14	Data analysis & SPC charts	Pandas, NumPy, Matplotlib
Python 3.14	Statistical testing	SciPy, Seaborn
Jupyter Notebook	Interactive development	VS Code extension
Microsoft Excel	Results reporting	Pivot Tables, Charts

2. Methodology

The analysis was structured into five sequential phases as shown in Figure 1.

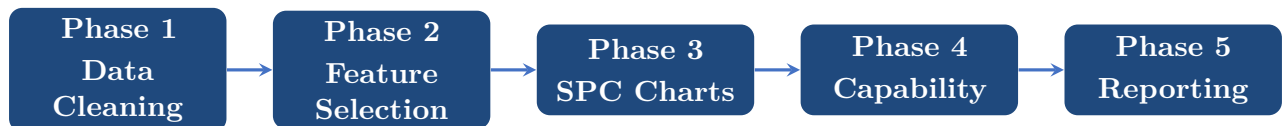


Figure 1: Five-Phase Analysis Methodology

2.1. Phase 1 — Data Cleaning and Preparation

Raw manufacturing data is rarely clean. The SECOM dataset is no exception — 91% of its columns contained missing values, and several parameters were missing data for over 90% of wafers.

The cleaning procedure applied was:

1. **Column dropping:** Any column with more than 50% missing values was removed. This is a standard threshold in industrial data analysis — parameters with insufficient data cannot support reliable statistical inference.
2. **Mean imputation:** For the columns retained, remaining missing values were filled with the column mean. This is an appropriate strategy for process parameters where values are expected to fluctuate around a stable central tendency.
3. **Label attachment:** Pass/fail labels from the separate `secom_labels.data` file were merged into the main dataframe.

The results of the cleaning phase are summarised in Table 3.

Table 3: Data Cleaning Summary

Metric	Before	After
Number of columns	590	562
Columns with missing data	538	0
Missing value count	>50,000	0
Columns dropped	—	28
Dataset shape	(1567, 590)	(1567, 563)

Key Finding

After cleaning, the dataset contained 1,567 wafers and 562 usable process parameters with zero missing values — ready for statistical analysis.

2.2. Phase 2 — Feature Selection

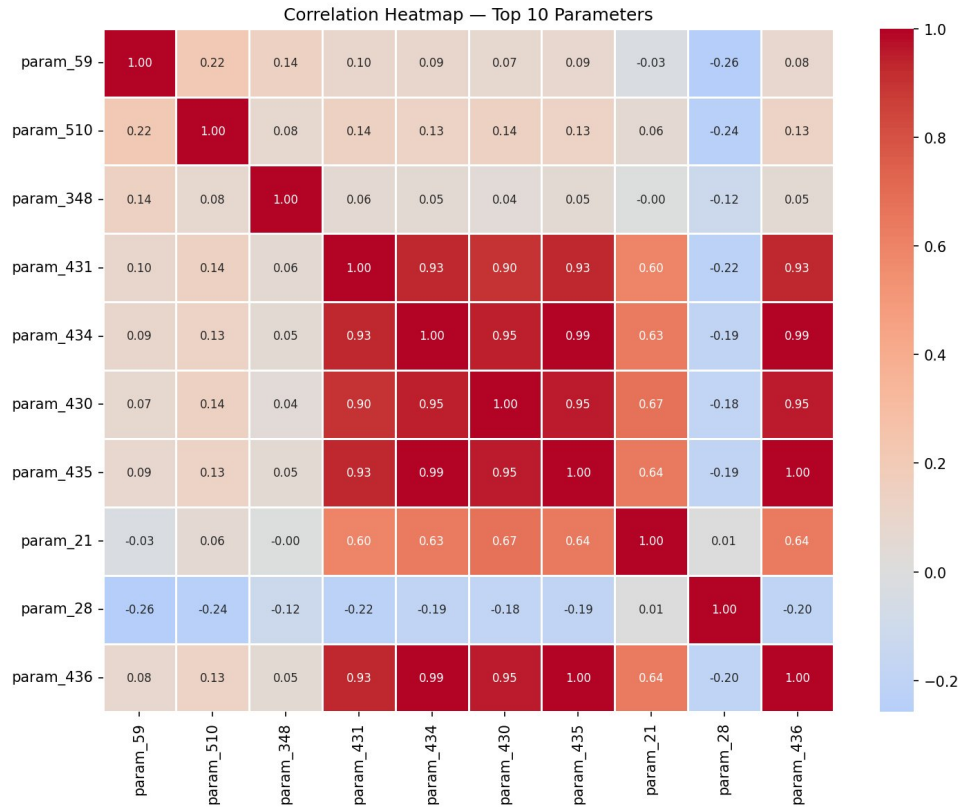
With 562 parameters remaining, direct SPC analysis of all columns would be computationally expensive and analytically meaningless. Two filtering steps were applied to identify the parameters most relevant to product quality.

2.2.1. Step 1: Variance Filtering

Parameters with a variance below 0.01 were removed. A parameter that barely changes across all wafers carries no discriminating information — it cannot explain differences between passing and failing units. This step removed **265 columns**, reducing the feature space from 562 to 297.

2.2.2. Step 2: Correlation Filtering

The absolute Pearson correlation between each remaining parameter and the pass/fail label was computed. The top 40 parameters — those most strongly associated with wafer failures — were retained for SPC analysis.

**Figure 2:** Correlation Heatmap of Top Process Parameters**Table 4:** Feature Selection Summary

Stage	Parameters Remaining
After data cleaning	562
After variance filter	297
After correlation filter	40

The top five parameters by correlation with wafer failure are listed in Table 5.

Table 5: Top 5 Parameters by Correlation with Wafer Failure

Rank	Parameter	Absolute Correlation
1	59	0.1558
2	510	0.1316
3	431	0.1203
4	434	0.1116
5	430	0.1096

Key Finding

The relatively low correlation values (0.08–0.16) are consistent with real semiconductor manufacturing data, where failures are typically caused by the complex interaction of many parameters rather than a single dominant factor.

3. SPC Analysis — Parameter 59

Parameter 59 was selected as the primary subject for detailed SPC analysis, as it exhibited the highest correlation with wafer failures ($r = 0.156$). Three control chart types were applied.

3.1. Control Chart Constants

All control charts were constructed using standard SPC methodology. For subgroup size $n = 5$, the relevant constants are:

$$A_2 = 0.577, \quad D_3 = 0, \quad D_4 = 2.114 \quad (1)$$

3.2. X-bar and R Control Charts

The dataset was divided into subgroups of size $n = 5$, yielding 313 subgroups. For each subgroup, the mean ($\bar{X}$) and range (R) were calculated.

Control limits were calculated as follows:

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \bar{R} = 2.96 + 0.577 \times 13.12 = 10.53 \quad (2)$$

$$LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \bar{R} = 2.96 - 0.577 \times 13.12 = -4.61 \quad (3)$$

$$UCL_R = D_4 \bar{R} = 2.114 \times 13.12 = 27.74 \quad (4)$$

$$LCL_R = D_3 \bar{R} = 0 \quad (5)$$

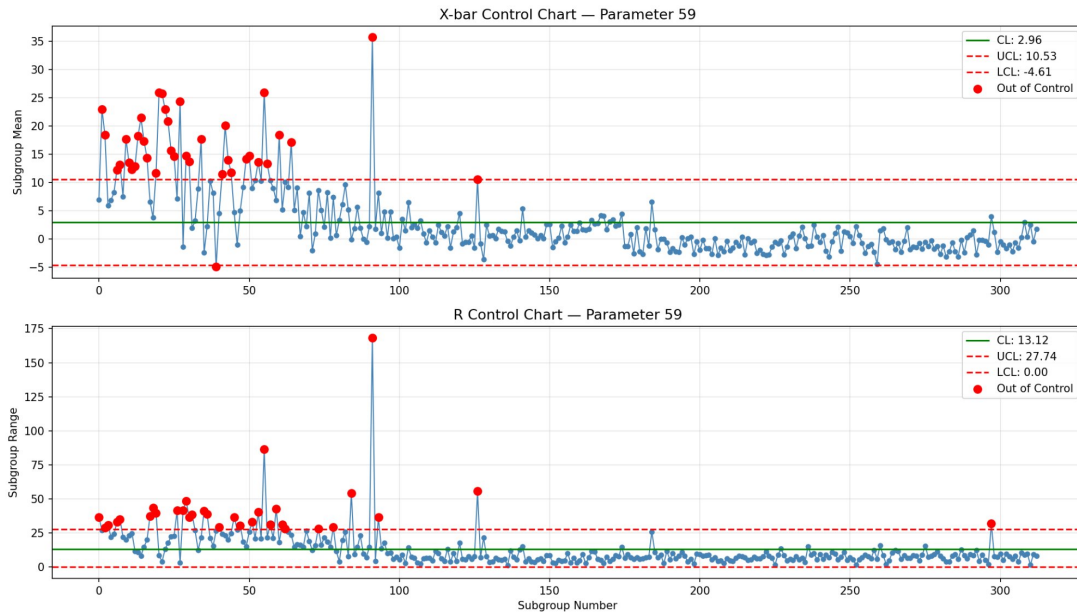


Figure 3: X-bar and R Control Charts for Parameter 59

Table 6: X-bar and R Chart Control Limits — Parameter 59

Chart	LCL	Centre Line	UCL
X-bar	-4.61	2.96	10.53
R	0.00	13.12	27.74

3.2.1. Out-of-Control Summary

Table 7: Out-of-Control Point Summary

Chart	OOO Points	% of Subgroups
X-bar	37	11.8%
R	32	10.2%

Key Finding

Over 10% of subgroups were out of control on both charts — a strong indicator of special-cause variation. Critically, the out-of-control points were heavily concentrated in the first 100 subgroups (approximately the first 500 wafers), after which the process stabilised significantly. This pattern suggests a **process shift event** occurred during early production — likely a machine adjustment, operator change, or material lot change.

3.3. EWMA Control Chart

The Exponentially Weighted Moving Average (EWMA) chart was constructed with smoothing parameter $\lambda = 0.2$ and control limit coefficient $L = 3$. Unlike the Shewhart chart which treats each subgroup independently, the EWMA weights all historical observations — giving more weight to recent data and less to older observations.

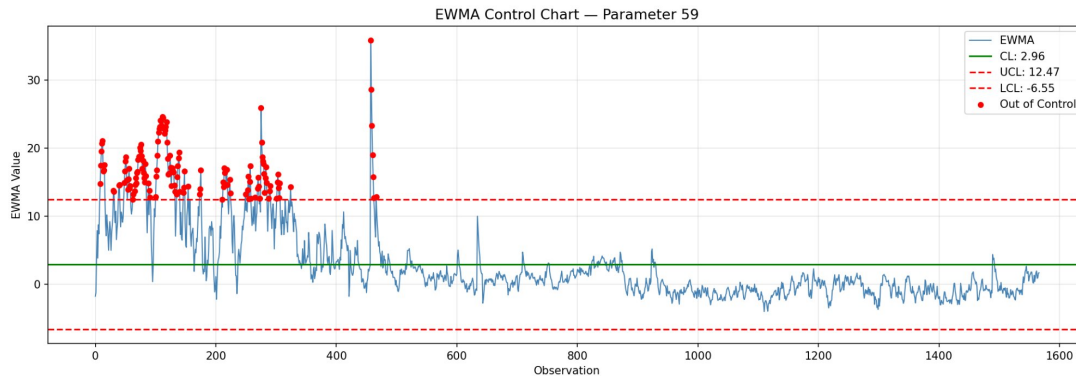


Figure 4: EWMA Control Chart for Parameter 59

Table 8: EWMA Chart Results — Parameter 59

Metric	Value
Smoothing parameter (λ)	0.2
Out-of-control points	143

Key Finding

The EWMA chart flagged 143 out-of-control observations — significantly more than the X-bar chart's 37. This is expected: the EWMA's memory makes it far more sensitive to sustained process shifts. The high OOC count confirms that the early production instability was not a series of isolated spikes, but a **persistent, sustained deviation** from the process target.

3.4. CUSUM Control Chart

The Cumulative Sum (CUSUM) chart is more sensitive than the Shewhart X-bar chart to small, sustained shifts in the process mean. The CUSUM chart for Parameter 59 confirmed the findings from the X-bar chart:

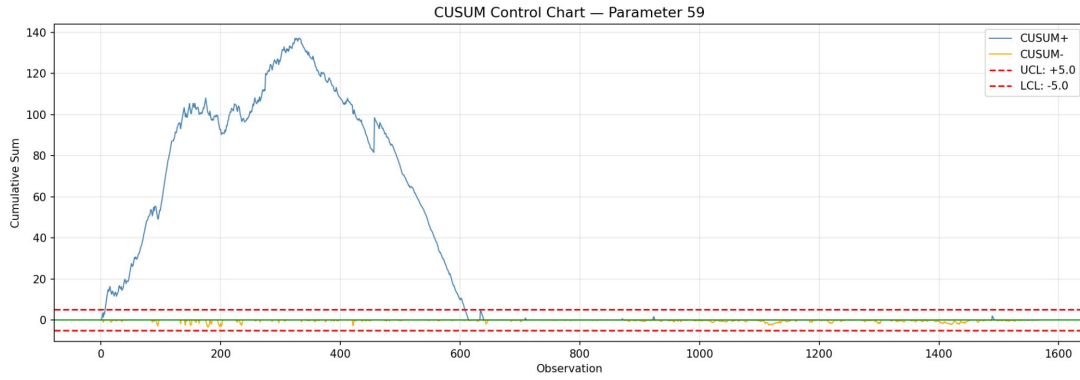


Figure 5: CUSUM Control Chart for Parameter 59

- CUSUM^+ values climbed steadily from observation 0 to approximately observation 330, reaching a cumulative sum of ≈ 137 — far exceeding the decision threshold of $+5$.
- A sharp reversal occurred between observations 330 and 620, indicating the process mean shifted downward during this period.
- From observation 620 onwards, CUSUM values remained near zero, confirming process stabilisation.

Key Finding

The CUSUM chart reveals that Parameter 59 experienced a **sustained upward drift** during the first third of production before correcting. This type of gradual shift is invisible to standard Shewhart charts but is clearly detected by CUSUM — demonstrating the value of using complementary chart types.

4. Process Capability Analysis

4.1. Background

Process capability indices quantify how well a process meets its specification limits. Two indices are commonly used:

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma} \quad (6)$$

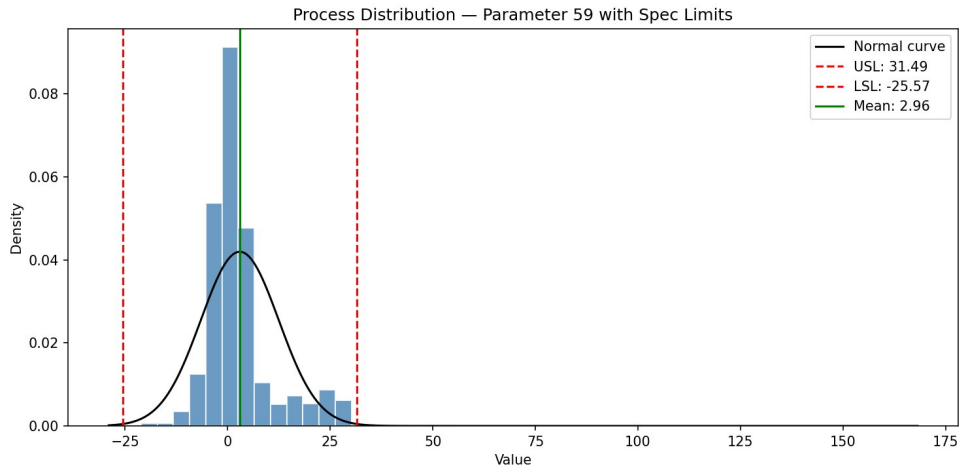
$$C_{pk} = \min \left(\frac{\text{USL} - \mu}{3\sigma}, \frac{\mu - \text{LSL}}{3\sigma} \right) \quad (7)$$

Where C_p measures potential capability (assuming the process is centred) and C_{pk} measures actual capability (accounting for process centring).

Table 9: Process Capability Index Interpretation

C_{pk} Value	Interpretation
≥ 1.67	Excellent — Six Sigma capable
$1.33 - 1.67$	Good — meets most industry standards
$1.00 - 1.33$	Adequate — some improvement needed
< 1.00	Incapable — process does not meet specifications

4.2. Results — Parameter 59

**Figure 6:** Process Capability Histogram for Parameter 59

Specification limits were estimated as $\pm 3\sigma$ from the process mean, yielding $USL = 31.49$ and $LSL = -25.57$. The capability indices were calculated as follows:

$$C_p = \frac{31.49 - (-25.57)}{6 \times 9.51} = 1.000 \quad (8)$$

$$C_{pk} = \min \left(\frac{31.49 - 2.96}{3 \times 9.51}, \frac{2.96 - (-25.57)}{3 \times 9.51} \right) = 1.000 \quad (9)$$

Table 10: Process Capability Results — Parameter 59

Metric	Value
Mean (μ)	2.96
Std Dev (σ)	9.51
USL	31.49
LSL	−25.57
C_p	1.000
C_{pk}	1.000
Assessment	Marginally Capable

Key Finding

With $C_{pk} = 1.000$, Parameter 59 sits exactly at the minimum acceptable threshold for process capability. While the process technically meets specifications, it provides zero margin for error — any further deterioration will result in out-of-specification output. Industry standard targets $C_{pk} \geq 1.33$.

4.3. Failure Rate Analysis — Early vs Late Production

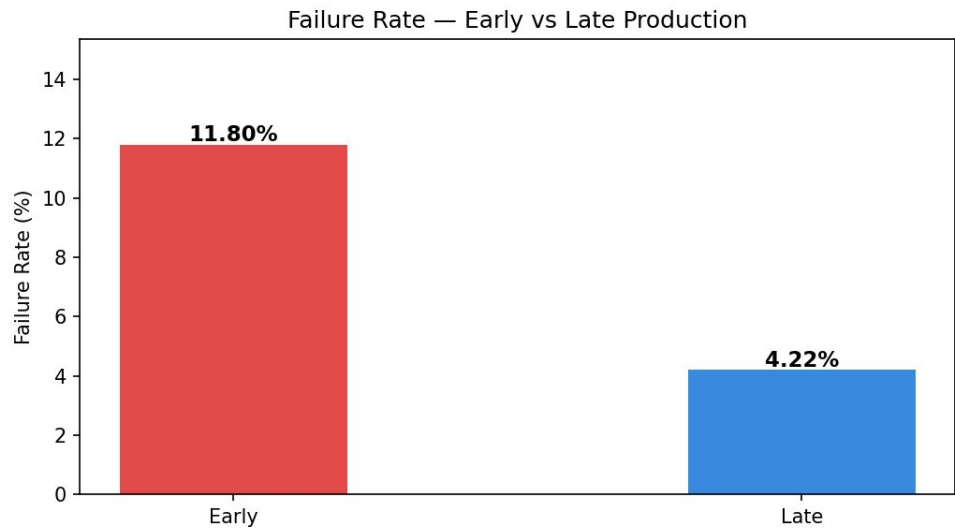


Figure 7: Failure Rate Comparison: Early vs Late Production

A critical finding emerged when the dataset was split into early production (first 500 wafers) and late production (remaining 1,067 wafers):

Table 11: Failure Rate by Production Period

Period	Wafers	Failure Rate
Early (wafers 1–500)	500	11.80%
Late (wafers 501–1567)	1,067	4.22%
Overall	1,567	6.60%

Key Finding

The failure rate during early production (11.80%) was **2.8× higher** than during late production (4.22%). This quantitatively confirms what the control charts showed visually — a significant process instability event during the first third of production. After the process stabilised, the failure rate dropped to less than half the overall average.

5. Key Findings and Recommendations

5.1. Summary of Findings

Table 12: Summary of Key Analytical Findings

Finding	Detail
Data Quality	91% of columns had missing values; 28 columns were dropped as unreliable. Mean imputation applied to remaining gaps.
Feature Reduction	590 parameters reduced to 40 analytically meaningful parameters through variance and correlation filtering.
Process Instability	Parameter 59 showed 37 OOC points (X-bar), 32 (R chart), and 143 (EWMA) — concentrated in the first 500 wafers.
Sustained Drift	CUSUM chart confirmed a sustained upward drift in Parameter 59 during early production, peaking around wafer 330 before correcting.
Process Capability	$C_p = C_{pk} = 1.000$ — marginally capable, with zero margin for further deterioration. Industry target is $C_{pk} \geq 1.33$.
Failure Rate Split	Early production failure rate (11.80%) was $2.8\times$ higher than late production (4.22%), directly confirming the instability period.
Failure Correlation	Wafer failures were weakly correlated with individual parameters ($\max r = 0.156$), suggesting multi-parameter interaction effects.

5.2. Multi-Parameter Interactions

As established in Phase 2 of the analysis, the highest individual Pearson correlation between any single process parameter and the final wafer pass/fail label was relatively low ($|r| = 0.1558$ for Parameter 59). This mathematical reality is highly characteristic of modern semiconductor manufacturing: defects are rarely the result of a single parameter catastrophically failing. Instead, they typically arise from complex, subtle interactions between multiple process variables.

For example, a deposition temperature might be slightly above its target but well within its specification limit, while a corresponding chamber gas flow rate might be slightly below target. Individually, neither deviation is severe enough to trigger a univariate control chart alarm (such as an X-bar or CUSUM chart). However, their combined interaction may create a sub-optimal processing environment that leads to a wafer failure.

To address this, the following multivariate strategies are recommended as the immediate next steps for process improvement:

1. **Principal Component Analysis (PCA):** Given the high dimensionality of the dataset (even after reducing the feature space to 40 parameters), PCA should be applied. This will eliminate multicollinearity and reduce the data into a smaller set of orthogonal principal components that explain the vast majority of the overall process variance.
2. **Hotelling's T^2 Control Chart:** Instead of monitoring parameters individually in isolation, a Hotelling's T^2 chart should be deployed. This multivariate SPC tool monitors the mean vector and covariance matrix of the selected parameters (or the principal components) simultaneously. It is explicitly designed to detect shifts in the relationships *between* variables that univariate charts are entirely blind to.
3. **Machine Learning for Interaction Detection:** Tree-based machine learning algorithms, such as Random Forests or Gradient Boosted Trees (e.g., XGBoost), should be trained on the process data. Unlike simple Pearson correlation, these models are inherently capable of mapping non-linear relationships and multi-way interactions, providing a much more accurate hierarchy of root causes.

Key Finding

Relying solely on univariate SPC is insufficient for a process of this complexity. Transitioning to multivariate SPC (specifically PCA coupled with Hotelling's T^2 charts) is critical. This will provide a comprehensive, system-wide perspective on stability and significantly improve the detection of interaction-driven defects before they result in scrapped wafers.

5.3. Recommendations

Based on the SPC analysis, the following recommendations are made:

1. **Investigate the early production period.** The concentration of out-of-control points in the first 500 wafers warrants a root cause investigation. Potential causes include machine warm-up effects, raw material lot changes, or operator variation during shift startup.
2. **Implement real-time CUSUM monitoring for Parameter 59.** The CUSUM chart demonstrated its ability to detect gradual drift that Shewhart charts would miss. Deploying this chart in real-time would enable earlier intervention.
3. **Investigate multi-parameter interactions.** The low individual correlations with failure suggest that no single parameter drives defects in isolation. A multivariate SPC approach (e.g., Hotelling's T^2) is recommended as a next step.
4. **Establish formal specification limits.** Accurate C_p and C_{pk} analysis requires defined USL and LSL values. Working with process engineers to formalise these limits would enable a complete capability study.

6. Conclusion

This project demonstrates the application of Statistical Process Control methodology to a real-world semiconductor manufacturing dataset. Starting from 590 raw process

parameters with significant missing data, the analysis systematically cleaned the dataset, identified the most quality-relevant parameters, and applied three complementary control chart methods to characterise process behaviour.

The analysis revealed that Parameter 59 — the parameter most strongly associated with wafer failures — exhibited significant instability during early production, with over 10% of subgroups violating control limits on both the X-bar and R charts. The CUSUM chart further revealed a sustained upward drift in the process mean during the first third of production, invisible to standard Shewhart charts, before the process self-corrected around wafer 600.

These findings highlight both the power of SPC as an analytical tool and the complexity of real manufacturing data — where failures rarely have a single identifiable cause, and where multiple complementary methods are needed to paint a complete picture of process health.

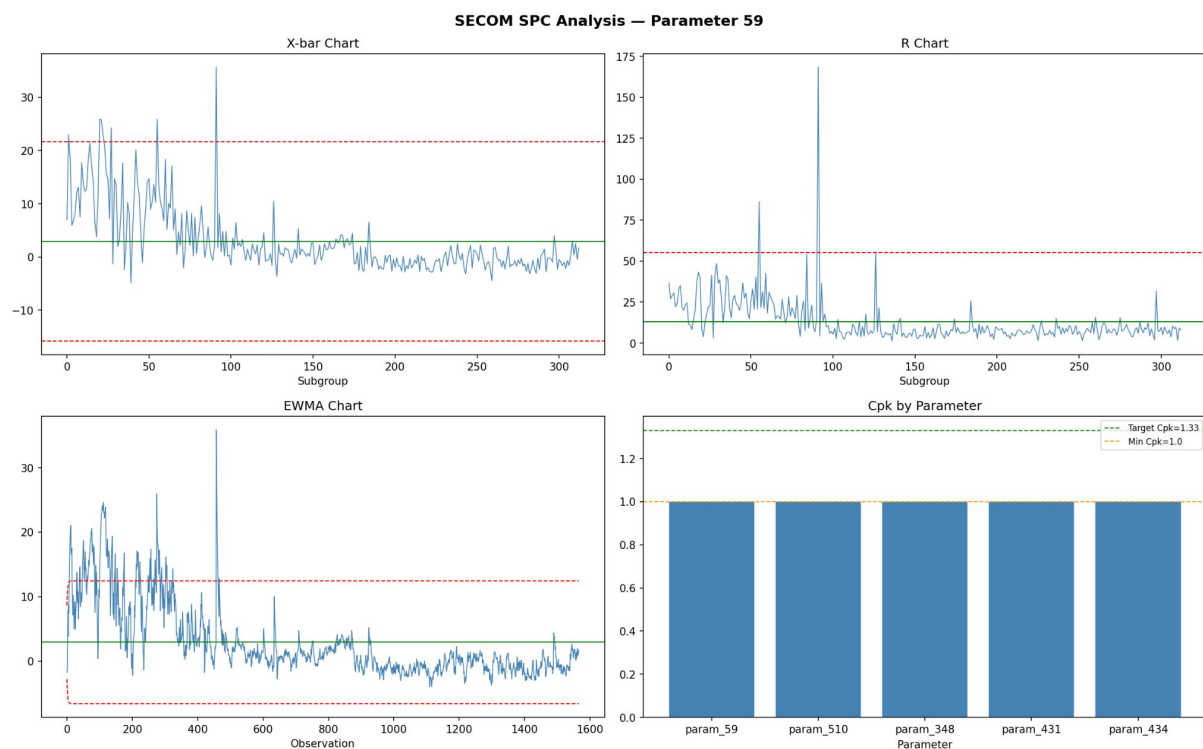


Figure 8: SPC Analysis Summary Dashboard

Project Deliverables:

- Python Jupyter Notebook — full analysis pipeline
- X-bar & R control charts (Parameter 59)
- CUSUM control chart (Parameter 59)
- EWMA control chart (Parameter 59)
- Microsoft Excel results summary dashboard
- This report

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